An empirical example: Four terminals and two areas

Daniel R. Miranda-Esquivel

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This is a worked example using a four terminals tree, distributed in two areas. Initially, all branches are equal.

```r
# Four taxa and two areas


## Load libraries

library(blepd)
```

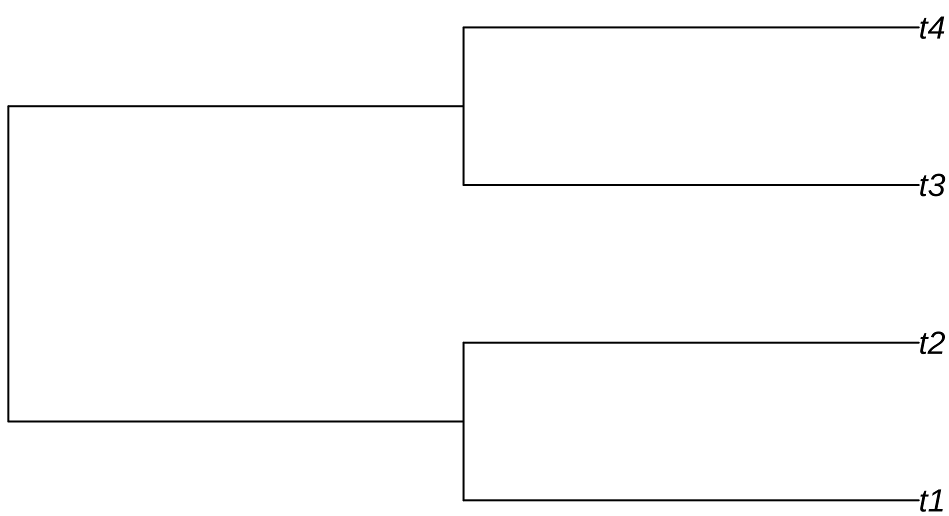
```
## Loading required package: ape
## Loading required package: picante
## Loading required package: vegan
## Loading required package: permute
## Loading required package: lattice
## This is vegan 2.6-8
## Loading required package: nlme
```

```r
library(ggtree)
```

```
## ggtree v3.14.0 Learn more at https://yulab-smu.top/contribution-tree-data/
##
## Please cite:
##
## Shuangbin Xu, Lin Li, Xiao Luo, Meijun Chen, Wenli Tang, Li Zhan, Zehan
## Dai, Tommy T. Lam, Yi Guan, Guangchuang Yu. Ggtree: A serialized data
## object for visualization of a phylogenetic tree and annotation data.
## iMeta 2022, 1(4):e56. doi:10.1002/imt2.56

##
## Attaching package: 'ggtree'
## The following object is masked from 'package:nlme':
##
##     collapse
## The following object is masked from 'package:ape':
##
##     rotate
```

```r
library(ggplot2)
```

```r
## Call the data
# Call the four terminals example
##data(package = "blepd")
## trees
data(tree)

## Plot the tree
plot(tree,use.edge.length =T)
```

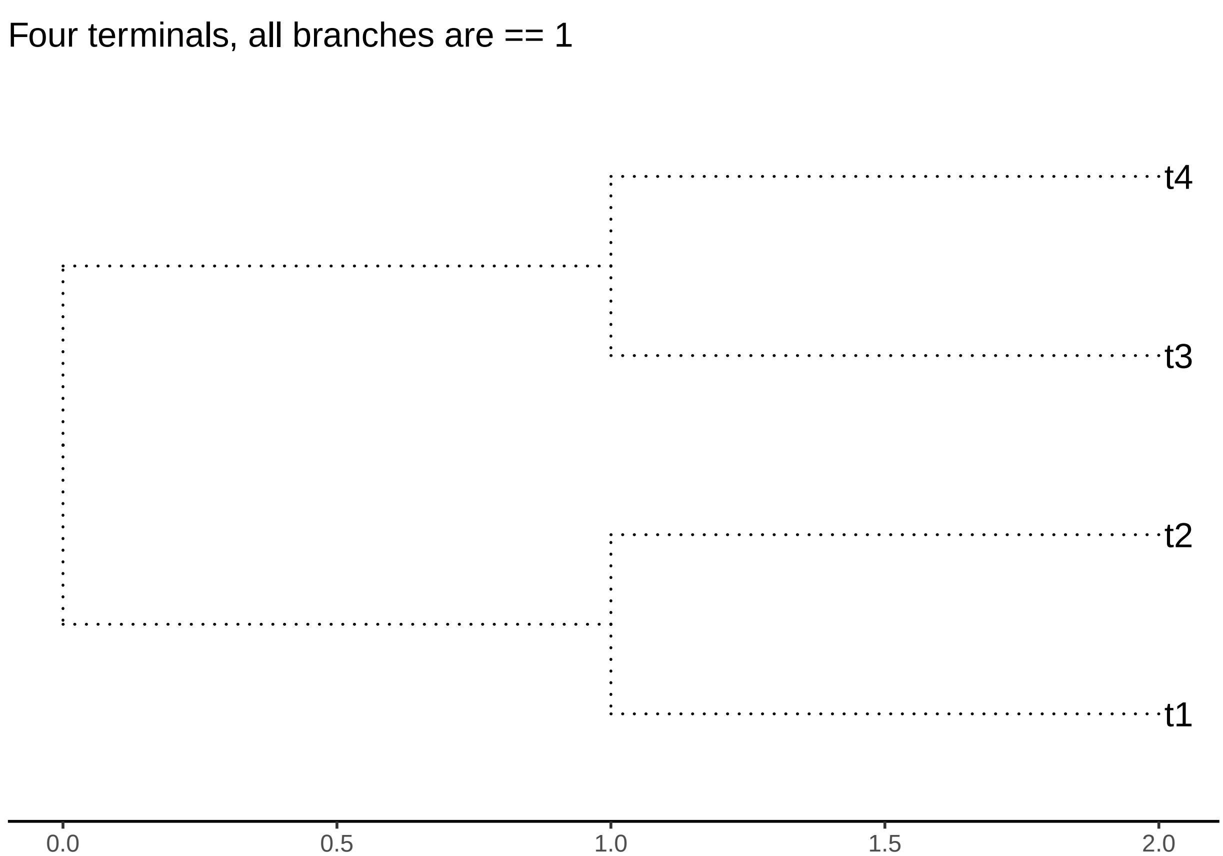



```r
initialTree <- tree

## Load and plot the distribution
data(distribution)
##str(distribution)
dist4taxa <- distribution
```

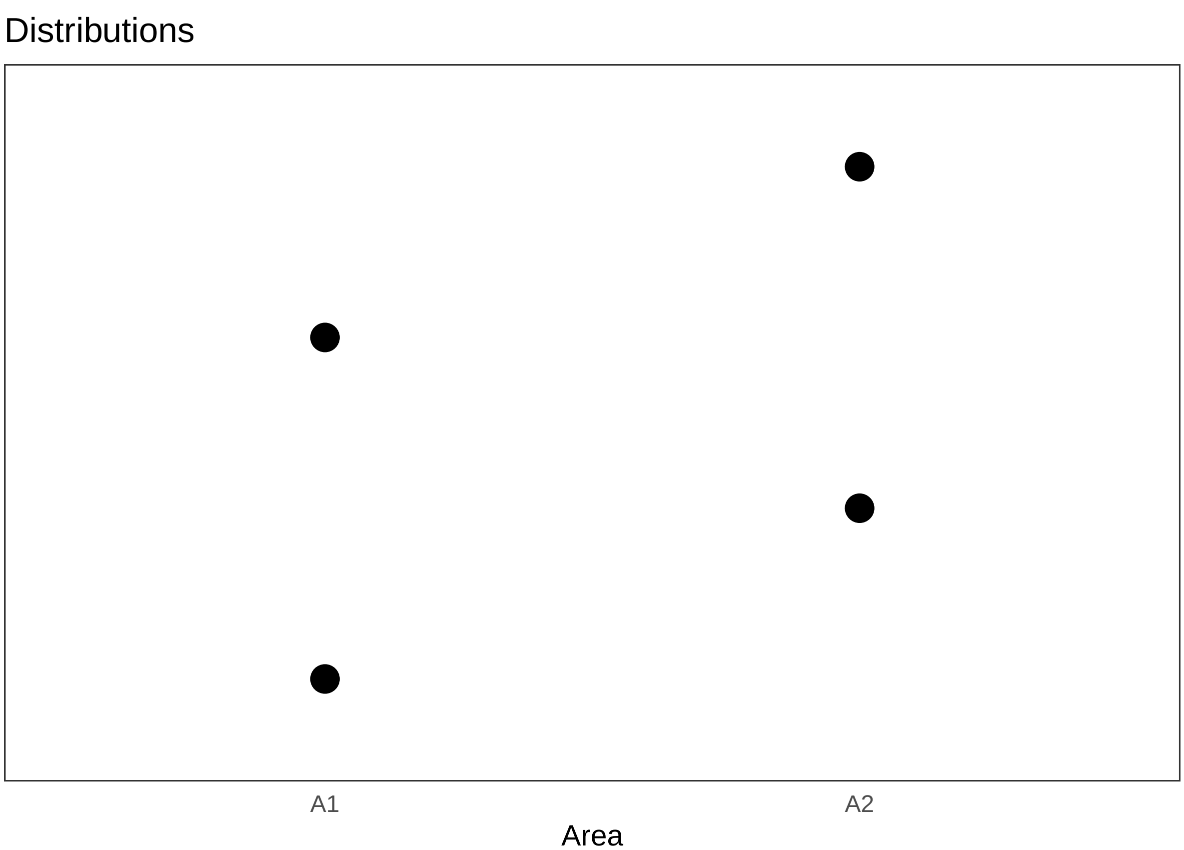
```r
## Change distribution to XY to be able to plot the data, otherwise, skip the next step

distXY <- matrix2XY(dist4taxa)

distXY
```

```
##   Terminal Area
## 1       t1   A1
## 2       t3   A1
## 3       t2   A2
## 4       t4   A2
```

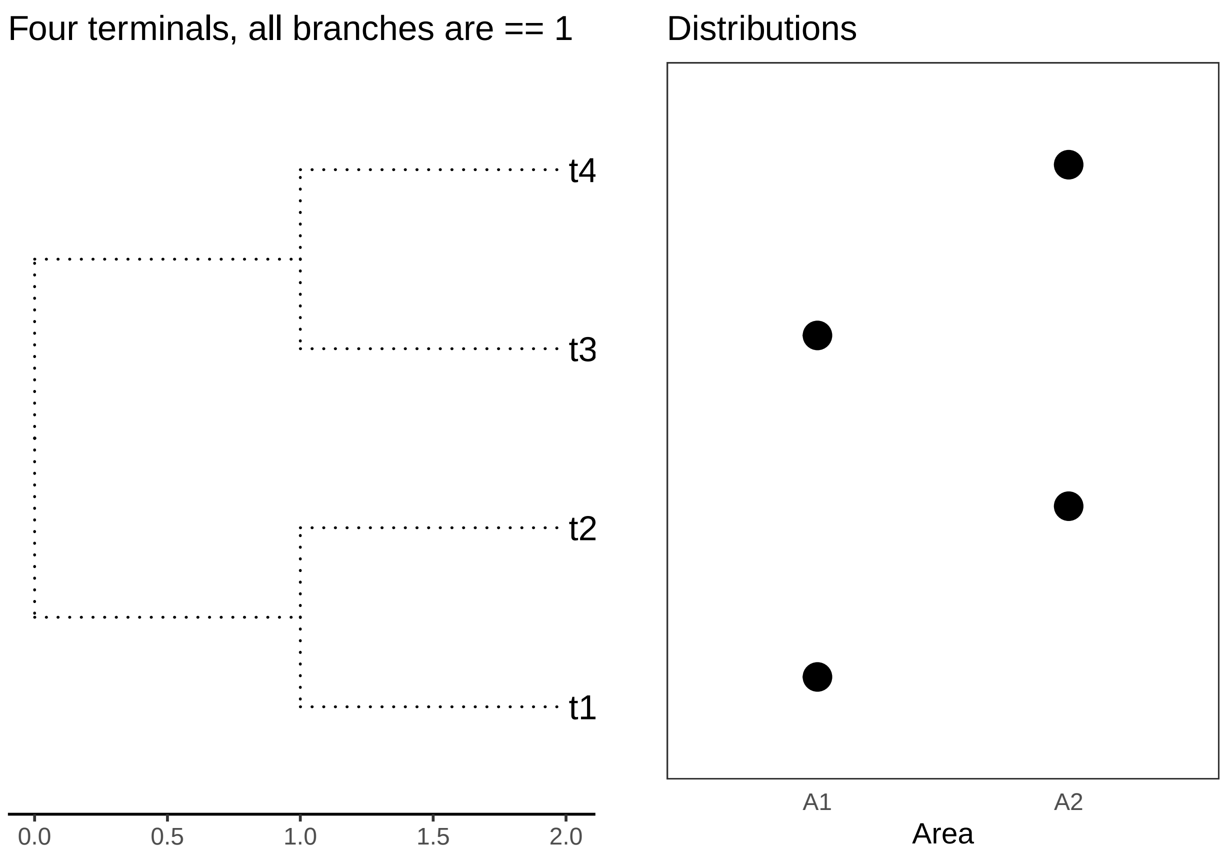
```r
## plotting

## the tree, using ggtree

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle("Four terminals, all branches are == 1")



print(plotTree)
```



```r
## The distribution using ggplot


plotDistrib <- ggplot(data=distXY,
                      aes(x= Area, y= Terminal , size =150))    +
               geom_point()      +
               theme_bw()                                       +
               theme( axis.text.y = element_blank(),
                      panel.grid.major = element_blank(),
                      panel.grid.minor = element_blank(),
                      axis.line = element_line(colour = NA_character_),
                      axis.ticks = element_blank(),
                      legend.position = "none" )               +
               labs(title = "Distributions",
                    y = "",
                    x = "Area")


##
print(plotDistrib)
```




```r
cowplot::plot_grid(plotTree, plotDistrib, ncol=2)
```




```
# Check whether names in both objects, initialTree and dist4taxa, are the same.


##all(colnames(dist4taxa) == initialTree$tip.label)

all(colnames(dist4taxa) %in% initialTree$tip.label)
```

```
## [1] TRUE
```

```
#Report the branch length, and calculate the PD values.


initialTree$edge.length
```

```
## [1] 1 1 1 1 1 1
```

```
initialPD <- PDindex(tree=initialTree, distribution = dist4taxa)

initialPD
```

```
## [1] 4 4
```

```
#As expected there is a tie between both areas.


#Now, we can see the impact of swapping the branch lengths; we will use the three
#available models to swap: "simpleswap","allswap","uniform", and we will swap the
#three classes of branches: "terminals","internals","all".
```

```r
##?swapBL


for( modelo in c("simpleswap","allswap","uniform") ){

        for( rama in c("terminals","internals","all") ){

 cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
            val <- swapBL(tree=initialTree,
                          distribution = dist4taxa,
                          model = modelo,
                          branch = rama
                         )
        printSwapBL(val, compact=TRUE)

         cat( "\n\n" )

     }

  }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
```

```
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1A2
## 1  100
```

```
#As expected, as all braches are EQUAL nothing happens


#Now, let us see if the results could be related to the branch length (or not).



## we could use a single command (*evalBranch*)


##?evalBranch

AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")


## The structure of the output

##str(AllTerminals)

## Print it as a friendly output

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    1        A1A2        A2   lower -1.024
## 2    2        A1A2        A1   lower -1.024
## 3    3        A1A2        A2   lower -1.024
## 4    4        A1A2        A1   lower -1.024
## 5    1        A1A2        A1   upper  1.062
## 6    2        A1A2        A2   upper  1.062
## 7    3        A1A2        A1   upper  1.062
```

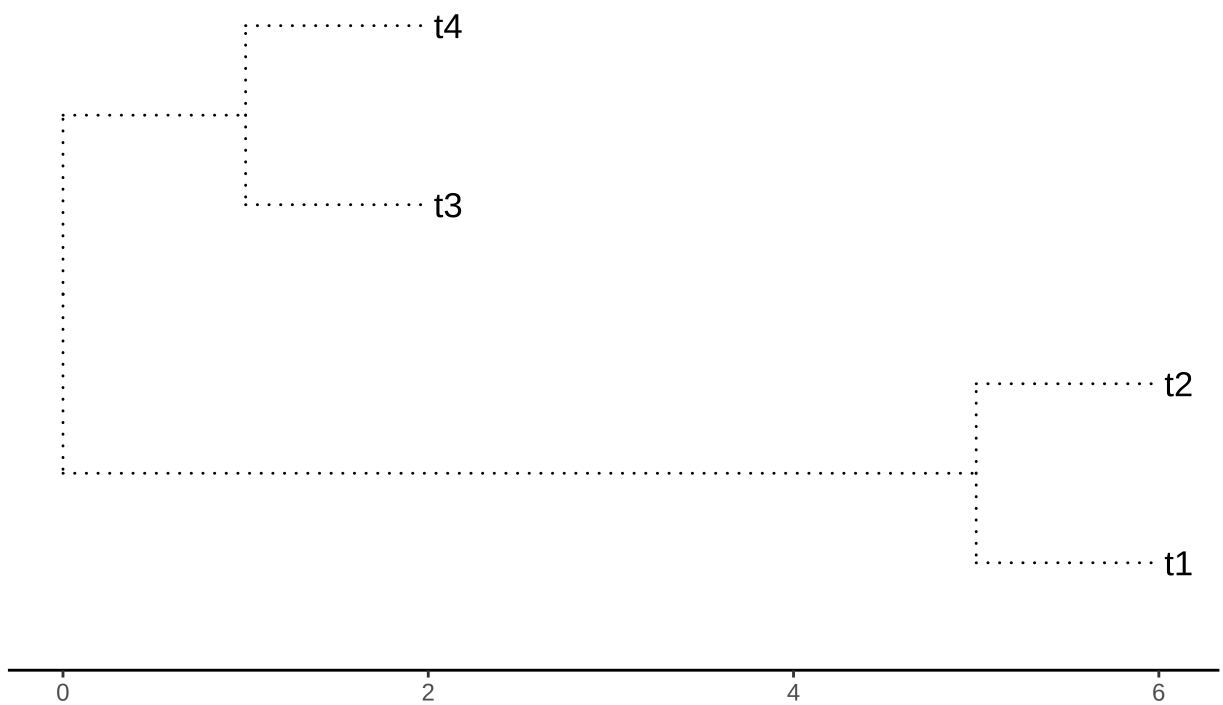
```
## 8    4         A1A2        A2   upper  1.062
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6        A1A2      A1A2   lower      0
## 2    7        A1A2      A1A2   lower      0
## 3    6        A1A2      A1A2   upper      0
## 4    7        A1A2      A1A2   upper      0
```

```r
#Now with a long branch


initialTree <- tree

initialTree$edge.length[1] <- 5

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=5. 2=1.




```
for( modelo in c("simpleswap","allswap","uniform") ){

      for( rama in c("terminals","internals","all") ){

         cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
          val <- swapBL(tree=initialTree,
                        distribution = dist4taxa,
                        model = modelo,
                        branch = rama
                       )
      printSwapBL(val, compact=TRUE)

       cat( "\n\n" )

    }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
```

```
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 11   77 12
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 27   38 35
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 50 50
```

```r
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

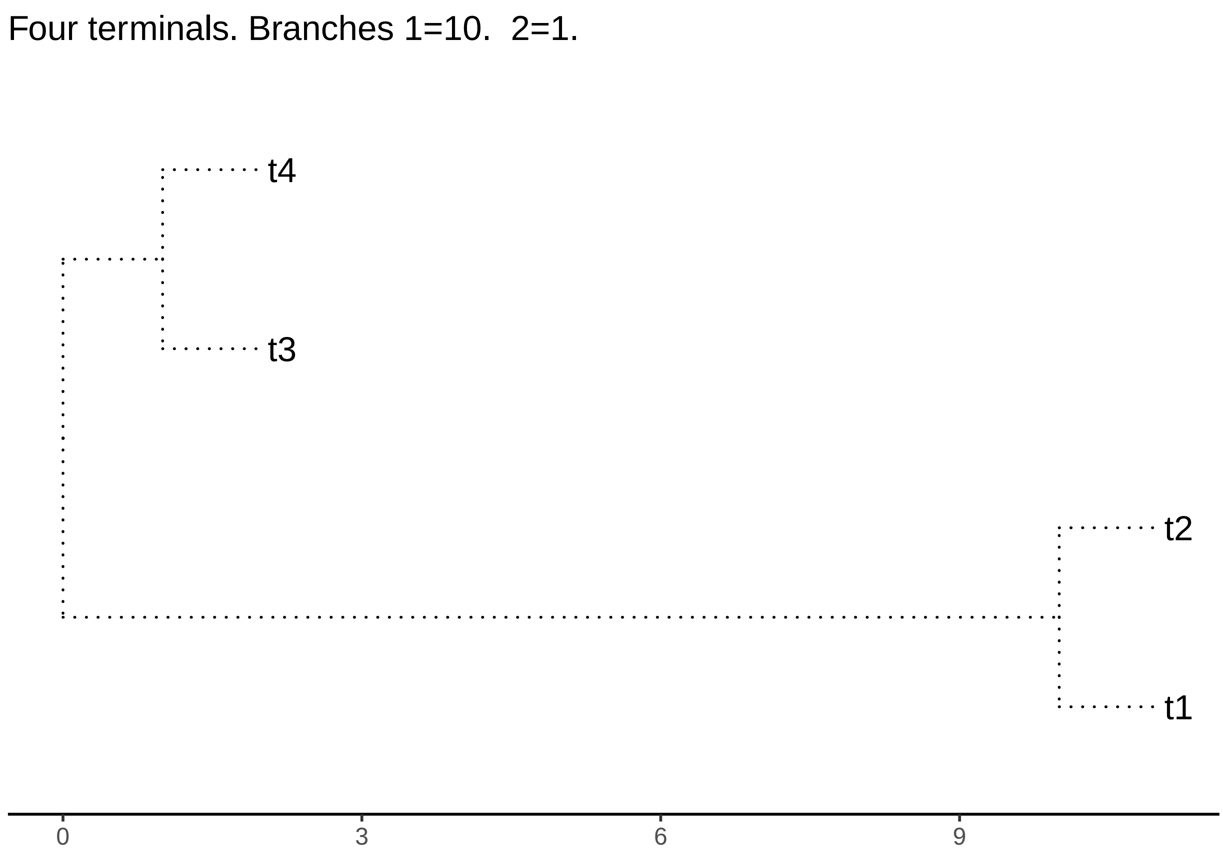
```
##   node initialArea FinalArea Aproach %Delta
## 1    1        A1A2        A2   lower -1.024
## 2    2        A1A2        A1   lower -1.024
```

```
## 3     3          A1A2         A2    lower -1.024
## 4     4          A1A2         A1    lower -1.024
## 5     1          A1A2         A1    upper  1.055
## 6     2          A1A2         A2    upper  1.055
## 7     3          A1A2         A1    upper  1.055
## 8     4          A1A2         A2    upper  1.055
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6        A1A2      A1A2   lower      0
## 2    7        A1A2      A1A2   lower      0
## 3    6        A1A2      A1A2   upper      0
## 4    7        A1A2      A1A2   upper      0
```

```r
# Now with a longer branch


initialTree <- tree

initialTree$edge.length[1] <- 10


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```




```
for( modelo in c("simpleswap","allswap","uniform") ){

      for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                        distribution = dist4taxa,
                        model = modelo,
                        branch = rama
                       )
      printSwapBL(val, compact=TRUE)

       cat( "\n\n" )

    }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
```

```
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 12   77 11
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 37   31 32
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 58 42
```

```r
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

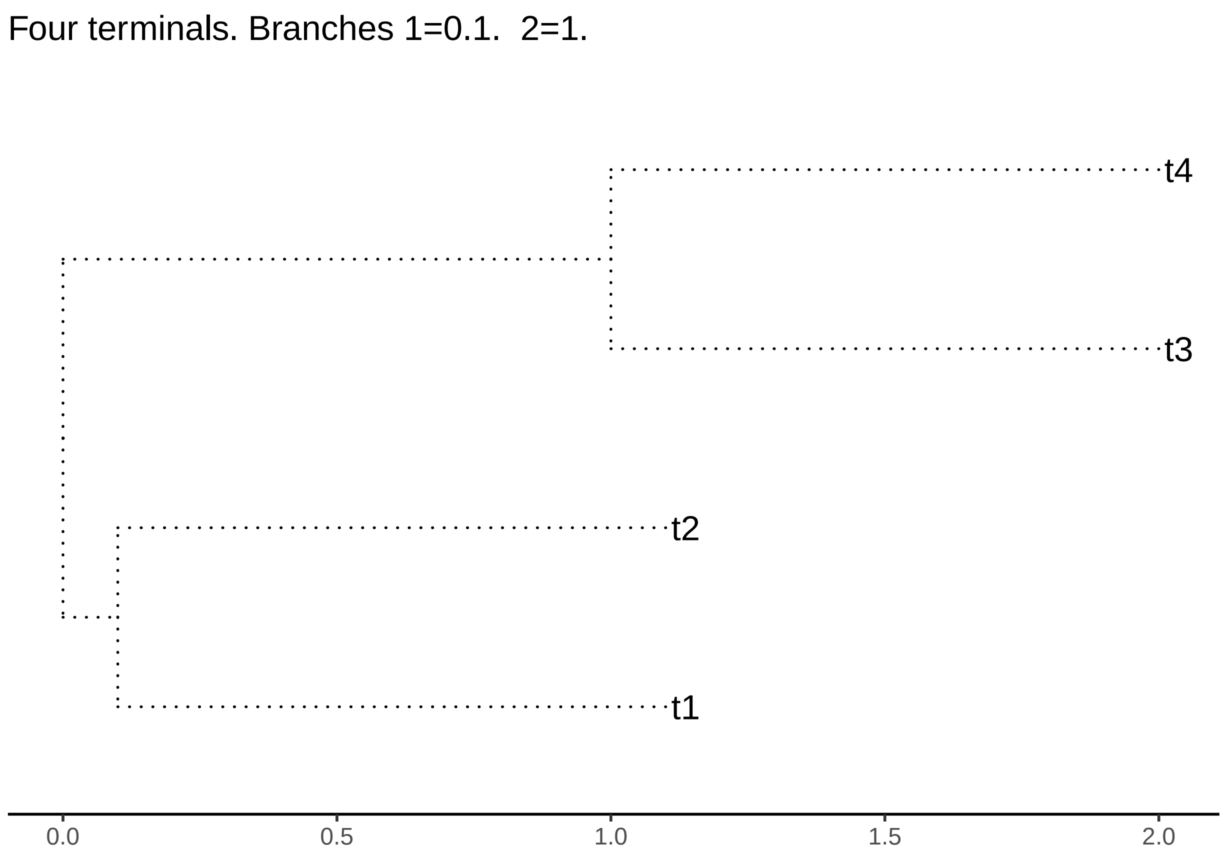
```
##   node initialArea FinalArea Aproach %Delta
## 1    1        A1A2        A2   lower -1.024
## 2    2        A1A2        A1   lower -1.024
```

```
## 3    3        A1A2        A2   lower -1.024
## 4    4        A1A2        A1   lower -1.024
## 5    1        A1A2        A1   upper  1.022
## 6    2        A1A2        A2   upper  1.022
## 7    3        A1A2        A1   upper  1.022
## 8    4        A1A2        A2   upper  1.022
```

```
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6        A1A2      A1A2   lower      0
## 2    7        A1A2      A1A2   lower      0
## 3    6        A1A2      A1A2   upper      0
## 4    7        A1A2      A1A2   upper      0
```

```
# With a short branch



initialTree <- tree

initialTree$edge.length[1] <- 0.1


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```




```
for( modelo in c("simpleswap","allswap","uniform") ){

      for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                        distribution = dist4taxa,
                        model = modelo,
                        branch = rama
                       )
      printSwapBL(val, compact=TRUE)

       cat( "\n\n" )

    }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
```

```
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 16   73 11
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 39   30 31
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##   A1A2
## 1  100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 48 52
```

```r
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    1        A1A2        A2   lower -1.024
## 2    2        A1A2        A1   lower -1.024
```

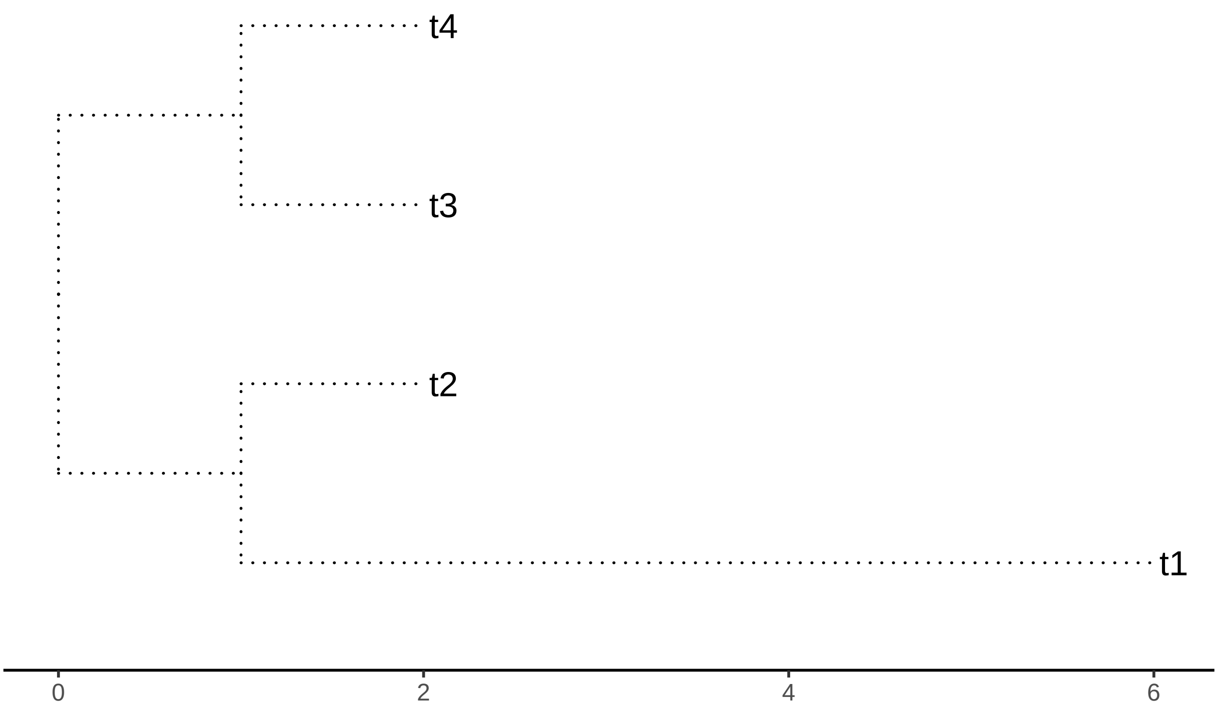
```
## 3    3        A1A2        A2   lower -1.024
## 4    4        A1A2        A1   lower -1.024
## 5    1        A1A2        A1   upper  1.051
## 6    2        A1A2        A2   upper  1.051
## 7    3        A1A2        A1   upper  1.051
## 8    4        A1A2        A2   upper  1.051
```

```
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6        A1A2      A1A2   lower      0
## 2    7        A1A2      A1A2   lower      0
## 3    6        A1A2      A1A2   upper      0
## 4    7        A1A2      A1A2   upper      0
```

```
#With a long branch in brach 2


initialTree <- tree

initialTree$edge.length[2] <- 5

plotTree <-  ggtree(initialTree, ladderize=TRUE,
                   color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=1. 2=5.




```
for( modelo in c("simpleswap","allswap","uniform") ){

      for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                         branch = rama
                        )
      printSwapBL(val, compact=TRUE)

       cat( "\n\n" )

    }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1 A2
## 1 63 37
##
##
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
```

```
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 77    8 15
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1 A2
## 1 48 52
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 37   29 34
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1 A2
## 1 49 51
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 51 49
```

```r
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach  %Delta
## 1    1          A1        A2   lower -80.193
## 2    2          A1        A1   lower       0
```

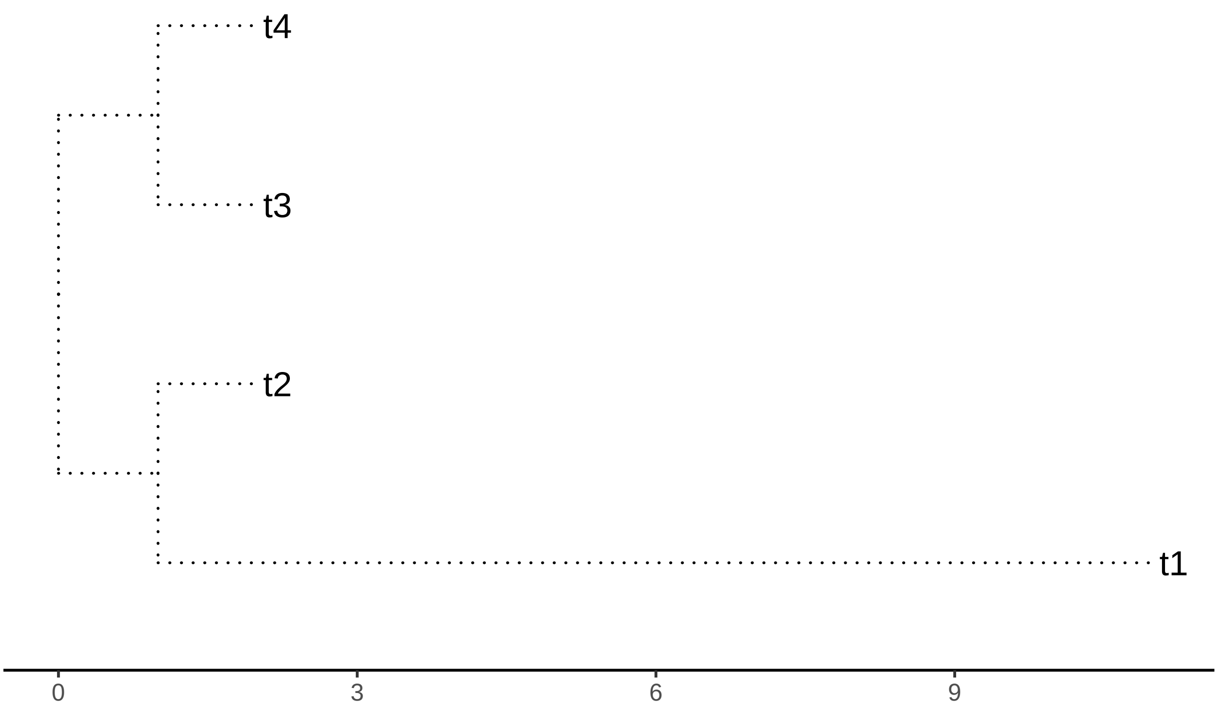
```
## 3    3          A1        A1   lower       0
## 4    4          A1        A1   lower       0
## 5    1          A1        A1   upper       0
## 6    2          A1        A2   upper 400.005
## 7    3          A1        A1   upper       0
## 8    4          A1        A2   upper 400.005
```

```
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6          A1        A1   lower      0
## 2    7          A1        A1   lower      0
## 3    6          A1        A1   upper      0
## 4    7          A1        A1   upper      0
```

```
# Now with a longer branch2


initialTree <- tree

initialTree$edge.length[2] <- 10


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=1. 2=10.




```
for( modelo in c("simpleswap","allswap","uniform") ){

      for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                         branch = rama
                        )
      printSwapBL(val, compact=TRUE)

       cat( "\n\n" )

    }

 }
```

```
##
##   Branchs swapped=   terminals.  model to test simpleswap reps 100
##   A1 A2
## 1 69 31
##
##
##
##   Branchs swapped=   internals.  model to test simpleswap reps 100
```

```
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 66   14 20
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1 A2
## 1 62 38
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 33   31 36
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1 A2
## 1 43 57
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##    A1
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 47 53
```

```
AllTerminals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "terminals",
                          approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach  %Delta
## 1    1          A1        A2   lower -90.096
## 2    2          A1        A1   lower       0
```

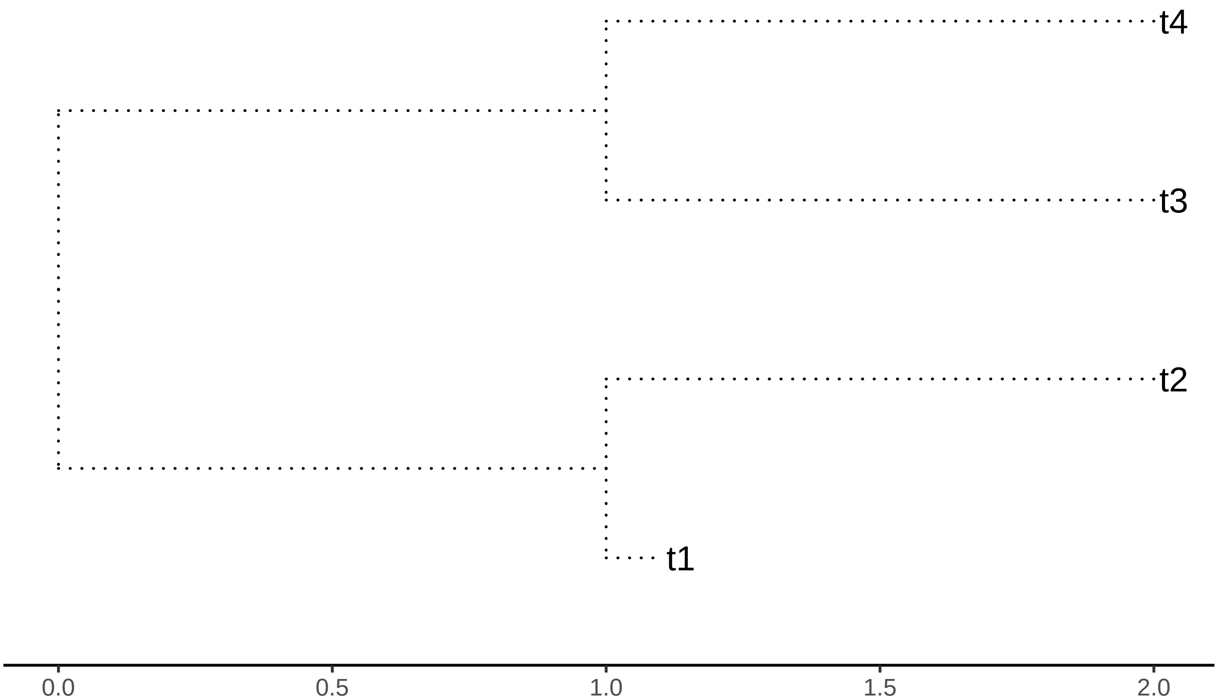
```
## 3    3           A1        A1    lower       0
## 4    4           A1        A1    lower       0
## 5    1           A1        A1    upper       0
## 6    2           A1        A2    upper 900.004
## 7    3           A1        A1    upper       0
## 8    4           A1        A2    upper 900.004
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6           A1        A1    lower      0
## 2    7           A1        A1    lower      0
## 3    6           A1        A1    upper      0
## 4    7           A1        A1    upper      0
```

```r
#now with a longer branch



initialTree <- tree

initialTree$edge.length[2] <- 0.1


plotTree <-  ggtree(initialTree, ladderize=TRUE,
                    color="black", size=0.5, linetype="dotted") +
             geom_tiplab(size=4.5, color="black") +
             theme_tree2() +
             ggtitle(paste("Four terminals. Branches 1=",initialTree$edge.length[1],
             ".  2=",initialTree$edge.length[2],".",sep=""))


##
print(plotTree)
```

Four terminals. Branches 1=1. 2=0.1.




```
for( modelo in c("simpleswap","allswap","uniform") ){

       for( rama in c("terminals","internals","all") ){

cat( "\n\t Branchs swapped=\t",rama,".\t",sep="")
           val <- swapBL(tree=initialTree,
                         distribution = dist4taxa,
                         model = modelo,
                         branch = rama
                        )
       printSwapBL(val, compact=TRUE)

        cat( "\n\n" )

    }

 }
```

```
##
##    Branchs swapped=   terminals.  model to test simpleswap reps 100
##    A1 A2
## 1 23 77
##
##
##
##    Branchs swapped=   internals.  model to test simpleswap reps 100
```

```
##    A2
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test simpleswap reps 100
##   A1 A1A2 A2
## 1 10   13 77
##
##
##
##   Branchs swapped=   terminals.  model to test allswap reps 100
##   A1 A2
## 1 50 50
##
##
##
##   Branchs swapped=   internals.  model to test allswap reps 100
##    A2
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test allswap reps 100
##   A1 A1A2 A2
## 1 29   38 33
##
##
##
##   Branchs swapped=   terminals.  model to test uniform reps 100
##   A1 A2
## 1 47 53
##
##
##
##   Branchs swapped=   internals.  model to test uniform reps 100
##    A2
## 1 100
##
##
##
##   Branchs swapped=   all.    model to test uniform reps 100
##   A1 A2
## 1 45 55
```

```r
AllTerminals <- evalBranch(tree=initialTree,
                           distribution = dist4taxa,
                           branchToEval = "terminals",
                           approach="all")

printEvalBranch(AllTerminals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach  %Delta
## 1    1          A2        A2   lower       0
## 2    2          A2        A1   lower -90.115
```

```
## 3    3          A2        A2   lower       0
## 4    4          A2        A1   lower -90.115
## 5    1          A2        A1   upper 900.408
## 6    2          A2        A2   upper       0
## 7    3          A2        A1   upper  90.005
## 8    4          A2        A2   upper       0
```

```r
## Now the internals

AllInternals <- evalBranch(tree=initialTree,
                          distribution = dist4taxa,
                          branchToEval = "internals",
                          approach="all")


printEvalBranch(AllInternals, compact=FALSE)
```

```
##   node initialArea FinalArea Aproach %Delta
## 1    6          A2        A2   lower      0
## 2    7          A2        A2   lower      0
## 3    6          A2        A2   upper      0
## 4    7          A2        A2   upper      0
```